RAHUL DAHIYA

Data & BI Analyst | Python, SQL, Power BI | GenAI Applications

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SUMMARY

Data and BI analyst with 2+ years across supply chain and transportation analytics. I build Python and SQL pipelines, Power BI models, and API automations over 1 lakh+ records of SAP and operational data, and at a multi-plant manufacturer these replaced manual daily reporting with automated data reporting used in management reviews. Now applying the same data foundation to GenAI systems: RAG, text-to-SQL, and tool-calling agents built with FastAPI and MongoDB.

EXPERIENCE

Business Intelligence Executive | Servotech Renewable Power System Ltd. Jun 2026 – Present

Supply chain and logistics analytics for a multi-plant manufacturer

- Replaced manual spreadsheet consolidation with a Python ETL pipeline that loads SAP billing and sales exports and operational trackers into a SQL database, standardising 1 lakh+ dispatch, billing, and freight records into one source for reporting and reconciliation.
- Own dispatch data and reporting end to end, delivering daily planned-vs-actual, same-day freeze plan, and next-day plan reports that senior management uses to track dispatch performance and follow up on held orders.
- Removed daily manual data entry from the dispatch approval process by automating it with Google Apps Script and the Sheets, Drive, and Gmail APIs, including rule-based live sync, multi-source lookups for missing vehicle, freight, and distance data, and auto-created evidence folders.
- Rebuilt the Power BI dispatch model (Power Query, DAX) on ~20,000 rows of dispatch history with live source connections, giving management one view of dispatch value, freight cost per order, transporter performance, and plan adherence.
- Standardised dispatch KPI definitions, including on-plan versus rescheduled dispatch with holiday rules and hold-reason tracking, so plant and sales teams report against the same numbers.
- Led data reconciliation for a full finished-goods store audit, matching physical counts across racks, floors, and outdoor storage to system stock and delivering an item-wise variance report signed off by store management.
- Leading the design of an in-house Transport Management System and supporting management in evaluating a commercial TMS vendor.

Business Analyst | BT Data and Surveying India Pvt. Ltd. Oct 2024 – May 2026

Transportation analytics and client reporting

- Cut manual effort in client reporting by automating preprocessing and validation of large-scale traffic count data with Python and SQL.
- Built ETL pipelines that cleaned, transformed, and standardised high-volume transportation datasets, giving analysts consistent inputs across client projects.
- Improved data quality and reporting accuracy by designing rule-based anomaly detection checks that flagged bad records before delivery.
- Developed Power BI dashboards for traffic trends, KPI monitoring, and peak-hour analysis used in client deliverables.
- Built trend forecasting models on traffic patterns to support transportation planning studies.

Software Engineer Intern | Tata Elxsi, Bengaluru Feb 2024 – Jun 2024

Data engineering and analytics

- Gave operations teams live data visibility by designing and deploying REST APIs and real-time dashboards.
- Reduced query execution time under high load by optimising SQL queries and backend workflows.
- Contributed to containerised data processing pipelines using Docker and Kubernetes.

PROJECTS

Plant Logistics Automation Suite Ongoing

Google Apps Script, Google Sheets, HTML web app, Power BI, SAP exports

- Designing an in-house TMS for a plant handling 10–30 dispatches a day, covering the flow from SAP sales order and stock and payment checks through vehicle placement, invoicing, LR, and e-way bill.
- Built a transporter e-bidding module that sends trip details to selected transporters by email and one-click WhatsApp, collects quotes through a web form, and shortlists the lowest bid, replacing manual freight requests over WhatsApp.
- Engineered around SAP GUI-only access (no API or OData) using structured exports and a central transporter master, on zero-cost Google Workspace tooling.

Enterprise Knowledge Assistant (production RAG with evaluation) GitHub

Python, OpenAI / Claude API, MongoDB Atlas Vector Search, BM25, FastAPI, Docker, GitHub Actions

- Built a cited-answer assistant over company policies and SOPs using hybrid retrieval (vector + BM25 keyword search, Reciprocal Rank Fusion, re-ranking) with metadata filters pushed into MongoDB Atlas $vectorSearch.
- Designed a 30-question evaluation suite covering hit rate, MRR, answer accuracy, faithfulness and refusal of unanswerable questions; the offline baseline scores 1.00 hit@5, 0.96 answer accuracy and 0% hallucination, enforced as a CI gate.
- Served through a streaming FastAPI API (SSE, file upload, API-key auth), containerised with Docker; 46 automated tests.

Text-to-SQL Analytics Agent GitHub

Python, PostgreSQL, SQLite, MongoDB, sqlglot, LLM tool-calling, FastAPI, Streamlit

- Built an agent that turns plain-English business questions into SQL, results, charts and a summary, with follow-up questions ("now split that by region") resolved from MongoDB conversation memory.
- Made it safe for real databases: sqlglot syntax-tree checks (SELECT-only, table allow-list, enforced LIMIT), read-only connections, statement timeouts, and LLM self-correction from database errors.
- Defined metrics once in a YAML semantic layer that also blocks double-counting joins; reached 24/24 execution accuracy against hand-written gold SQL on both SQLite and PostgreSQL; 71 tests.

AI Integration Agent (MCP server and REST API orchestration) GitHub

Python, MCP Python SDK, FastAPI, httpx, OAuth2, webhooks, SQLite, Docker

- Built a support copilot that resolves tickets end to end across order, helpdesk and email APIs, triggered by HMAC-signed webhooks with replay protection and de-duplication.
- Engineered a resilient API layer: API-key, OAuth2 client-credentials and bearer auth, token-bucket rate limiting, exponential backoff honouring Retry-After, and idempotency keys so a lost response never causes a double refund.
- Exposed the integrations as an MCP server (9 tools, resources, prompt) sharing one tool registry with customer-scoped permissions, human approval for high-risk actions and a PII-masked audit log; 59 tests.

Data & AI Lab (supply chain and operations analytics projects) GitHub

Python, pandas, NumPy, DuckDB, SQL, PyYAML, pytest, GitHub Actions

- Build each project over three days the way a team works: plan issue, pull request, code review, issues for findings, fixes with tests, CI, then merge.
- **Intermittent Demand Forecaster:** classifies spare-parts demand (ADI / CV^2), backtests Croston, SBA and TSB per SKU and sets reorder points at a target service level; cut error on intermittent SKUs from 0.80 to 0.63 RMSSE against a naive forecast; 69 tests.
- **SOP Retrieval Benchmark:** scores retrieval over site-specific warehouse SOPs (BM25, query expansion, site filter) on 315 labelled questions with recall@k, MRR and a wrong-site rate; showed a synonym glossary that lifted paraphrase recall@5 from 0.48 to 0.91 lowered it on held-out wording (0.70 to 0.60); 71 tests.
- **Rate Card History Loader:** loads an at-least-once CDC stream of carrier rate changes into an SCD Type 2 table in DuckDB with an LSN watermark, applied-event ledger and quarantine, then re-rates invoices at the rate valid on the ship date; a replayed batch changed nothing and all 21 planted billing errors were found with no false flags; 83 tests.

SKILLS

Programming & Data: Python (Pandas, NumPy), SQL, ETL pipelines, CDC and SCD Type 2 loading, data modelling, data contracts and data quality checks, EDA, Google Apps Script, Node.js

Databases: PostgreSQL, DuckDB, MongoDB (Atlas Vector Search), Snowflake

GenAI & ML: RAG, LLM tool-calling agents, MCP, prompt engineering, LLM evaluation, retrieval evaluation (BM25, recall@k, MRR), LangChain, FAISS, OpenAI and Claude APIs, scikit-learn, forecasting (including intermittent demand)

APIs & Deployment: FastAPI, Flask, REST API integration, Streamlit, Docker, Kubernetes, AWS, Git

BI & Reporting: Power BI, Power Query, DAX, Advanced Excel, Google Sheets, KPI design

Domain: SAP (SD/MM data extraction), dispatch planning, transport management, freight cost analysis, inventory audit; LLM training data work (rubric writing, response rating, safety evals)

EDUCATION

M.Sc. Computer Science (Data Analytics) 2022 – 2024

Kerala University of Digital Sciences, Innovation and Technology, Thiruvananthapuram